

# Session 3

Case Study: Singapore's Smart Nation, Open Government Products, and Singapore Government Tech Stack

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# SMART NATION

S I N G A P O R E

# Smart Nation journey



1963



1981



1997



2000



2007



2014



2016



2017



2018



2022



2023



2024

# Vision

- Singapore's **Smart Nation initiative**, led by the Ministry of Digital Development and Information (MDDI), aims to build a thriving digital future for everyone.
- Launched in **2014**, Smart Nation 1.0 sought to improve lives through seamless and meaningful use of technology.
- Since its launch, Singapore has integrated digital technologies deeply into daily life, creating new opportunities while also needing to manage risks to well-being and social cohesion.
- In **2024**, Singapore introduced **Smart Nation 2.0**, a refreshed vision focused on using technology more effectively to transform the nation's future.
- The new phase shifts from primarily building digital capabilities to applying them strategically to shape Singapore's long-term development.

# Goals

# Trust



Enhance the security and resilience of our digital infrastructure



Strengthen our fight against harmful online activities



Expand our trusted digital spaces together

# Growth



Power our digital economy



Empower our enterprises  
and workers



Prepare our next generation

# Community



Strengthen inclusion:  
leaving no one behind



Strengthen the unity of our  
community



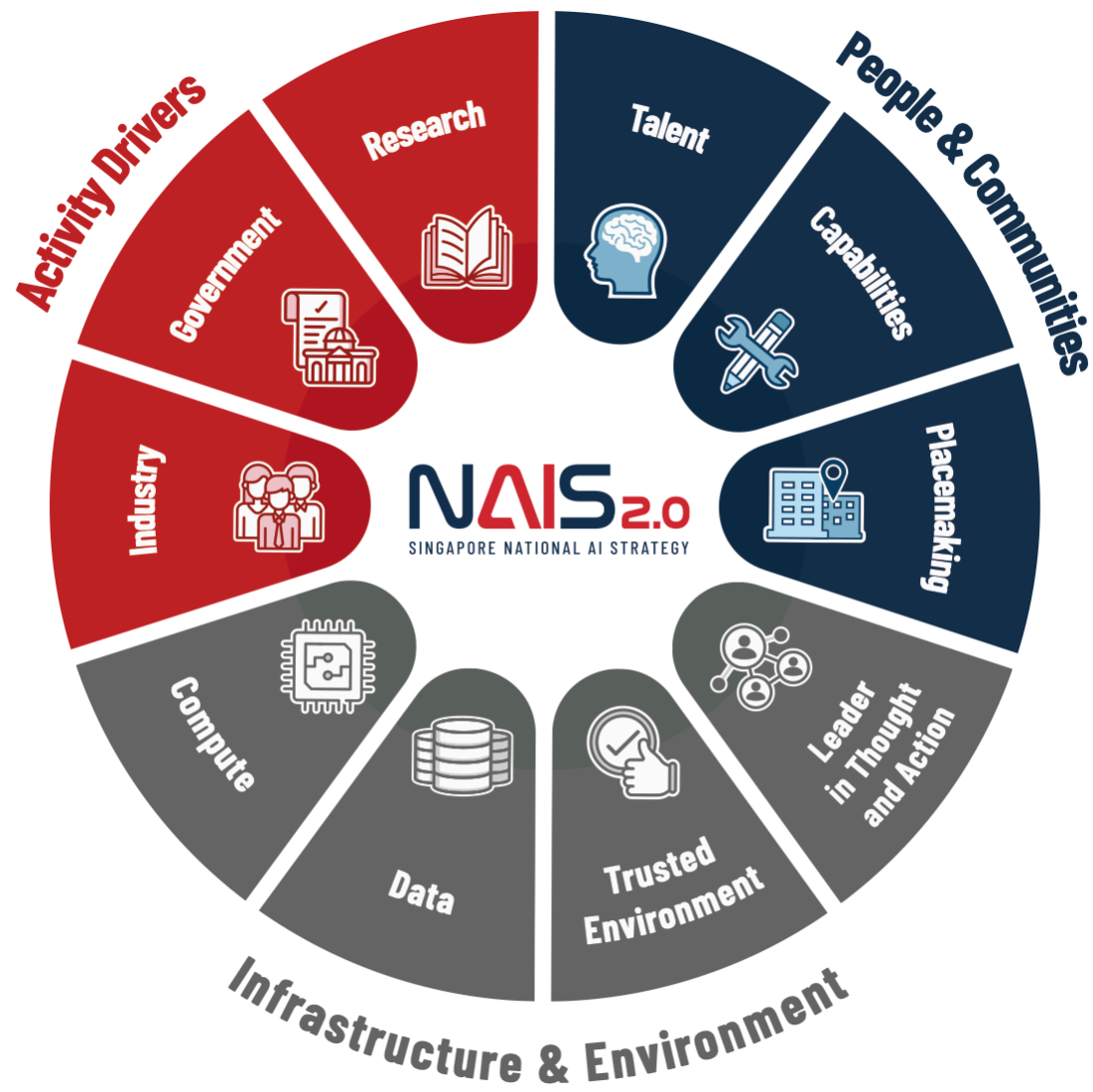
Strengthen partnerships  
with citizens



# Smart City achievements

- **99%** of all transactions between citizens and the government can be completed online.
- **High citizen and business satisfaction** with Government Digital Services at **83%** in 2023.
- **99%** of resident households are **connected to the internet**, as of 2022.
- **95%** of small and medium enterprises (SMEs) have adopted at least one digital technology from nine categories: cybersecurity, cloud, e-payment, e-commerce, data analytics, artificial intelligence, Internet of Things, blockchain, and immersive media.

# National AI Strategy

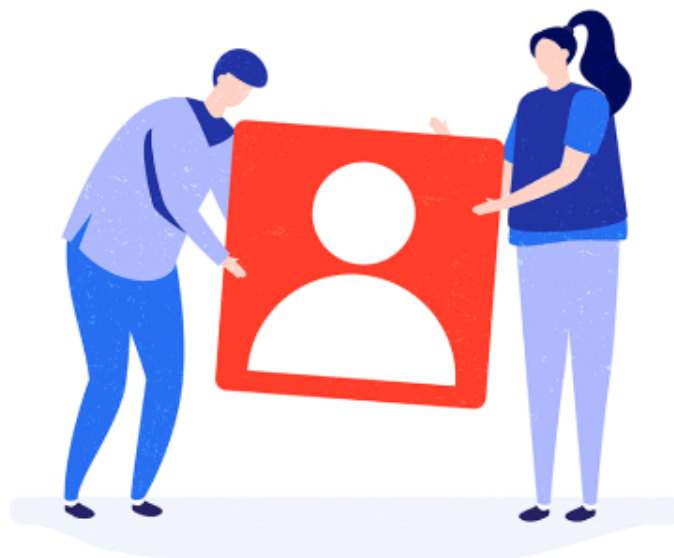


# Open Government Products

# Approach



Solve real problems



Build for the user



Push for change

# Products

- More than 40 products in progress
- Use **report cards** mechanism to provide a transparent overview of the work being done on each of the products

## **Tackle real problems**

Share why and how the product solves a problem with updates along the way.

## **Resource responsibly**

Product costs are made open as possible to stay accountable to taxpayers and citizens.

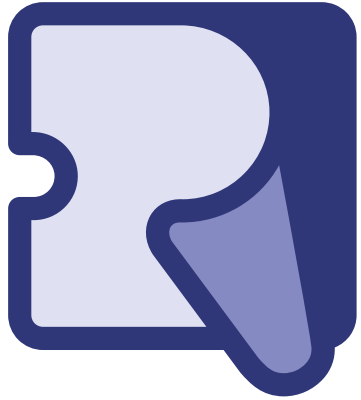
## **Track impact clearly**

Share product outcomes to ensure delivery of clear value to the public.



# SCAMSHIELD

- **ScamShield** helps users detect, check, and report suspicious messages, calls, and links.
- Reported scams are verified by authorities, helping them track and respond to new scam variants.



# REDEEMSG

- Paper voucher campaigns were slow to launch and difficult for residents to claim and use.
- Merchants experienced long delays in receiving payments.
- Existing digital alternatives were available but scattered across multiple complicated apps.
- **RedeemSG** provides a unified platform for government agencies and non-profits to create, distribute, and monitor vouchers nationwide.
- Citizens redeem vouchers and merchants accept them through the same system.



- Paper-based tracking made it difficult for organisations to monitor distributions or adapt plans in real time.
- This led to long waiting times, inaccurate records, and limited ability to reach more beneficiaries.
- **DistributeSG** enables government agencies and non-profits to run and monitor distribution campaigns with real-time tracking.
- It includes an admin portal for organisers to set up campaigns and a mobile app for on-the-ground tracking of distributed items.



# Initiatives



- **Build for Good** is a citizen engagement initiative by Open Government Products that encourages citizens to help improve Singapore.
- People can participate through a **hackathon**, an **accelerator for winning teams**, or by joining the **community**.
- The initiative aims to create a supportive space where individuals feel empowered to contribute positively, experiment, and build solutions that make the world better.



# hack for public good

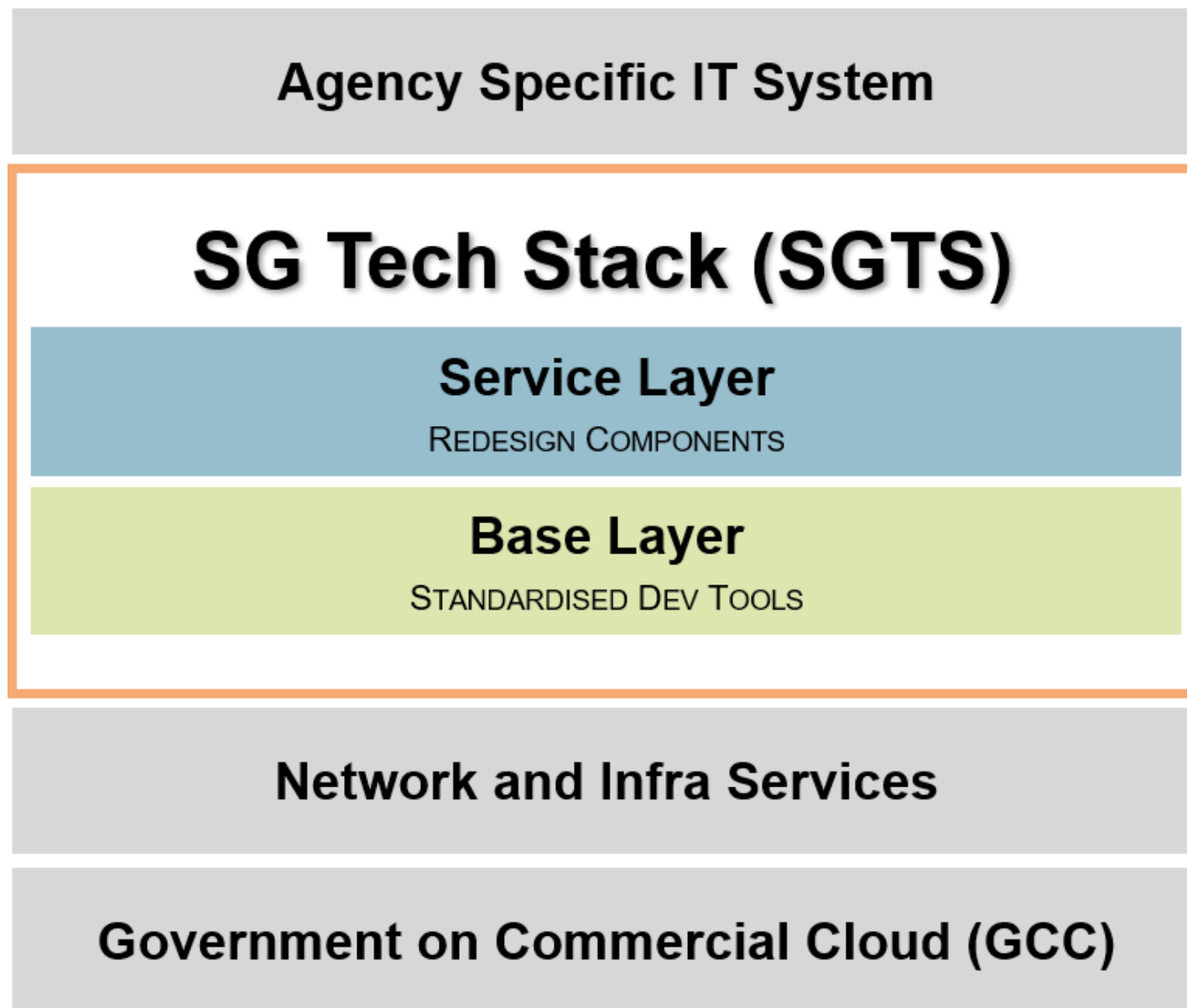
- **Hack for Public Good (HFPG)** replaces top-down product development with a community-driven approach.
- Instead of leadership setting the direction, tech teams engage directly with the community to identify public good problems in Singapore.
- These teams then design and build solutions based on what they discover.

# Singapore government tech stack

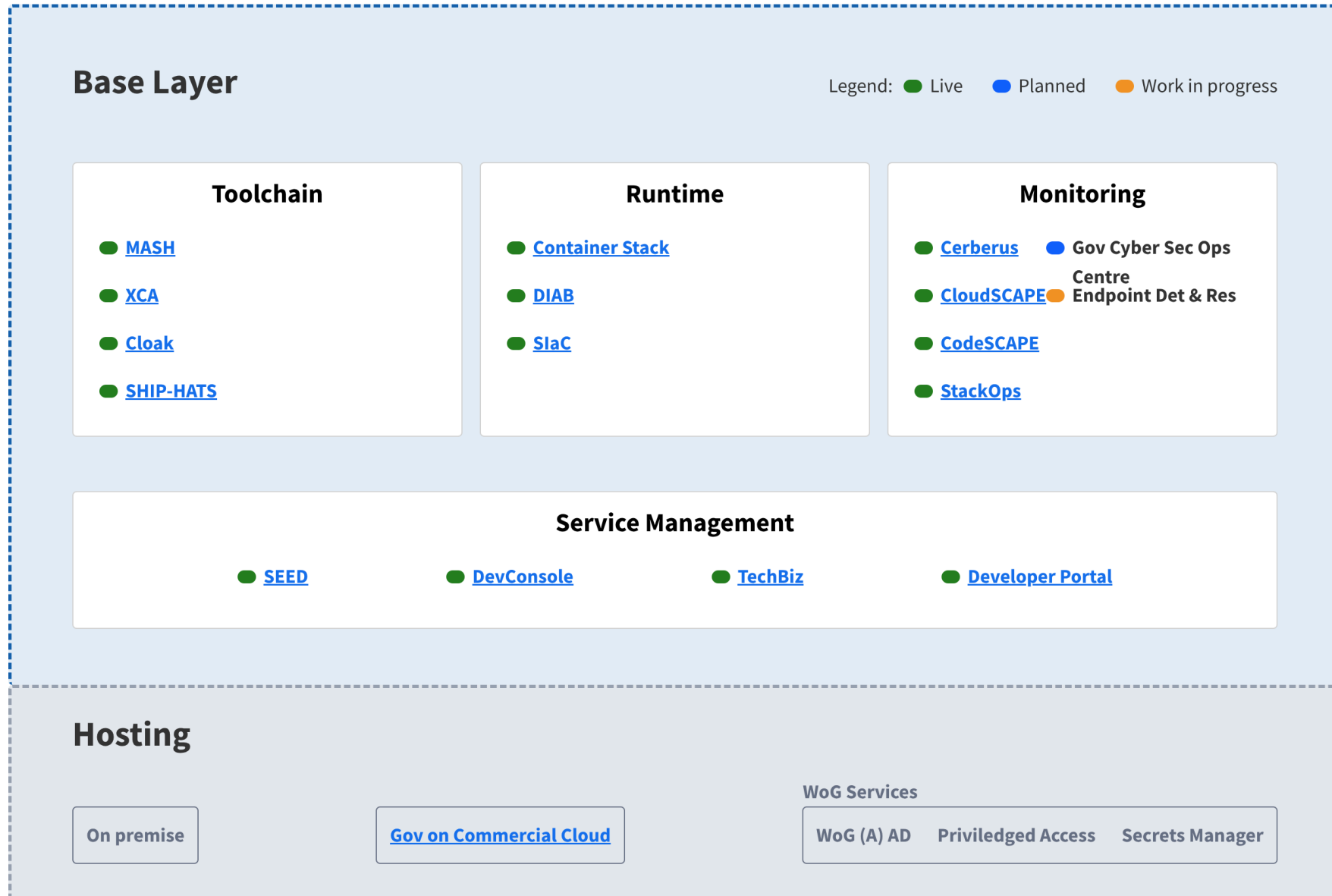
# Overview

- The **Singapore Government Tech Stack (SGTS)** modernises development practices across the whole government.
- The goal is to support the creation of secure, responsive systems that keep up with evolving citizen and business needs.

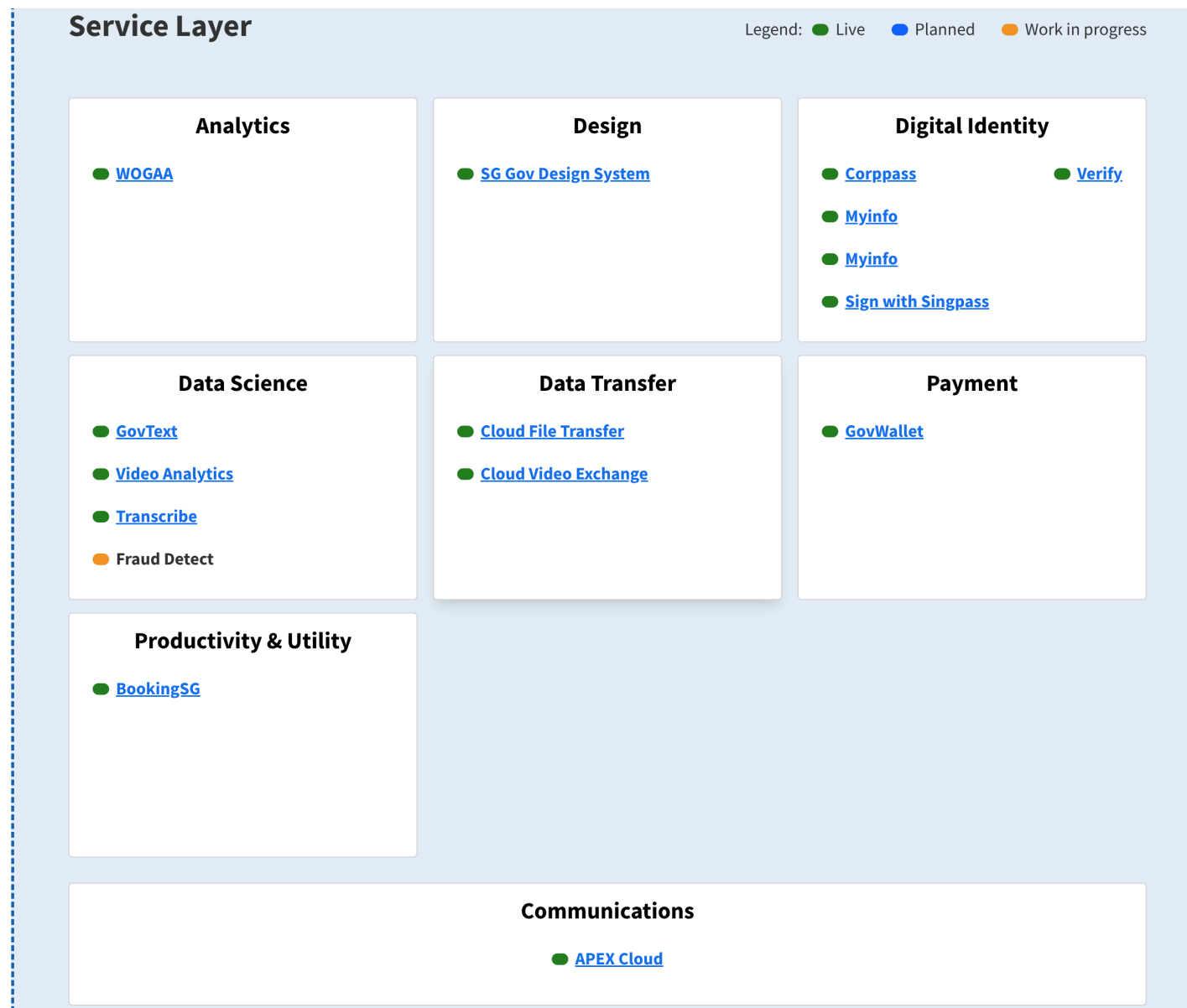
# Architecture



# Architecture - base layer



# Architecture - service layer



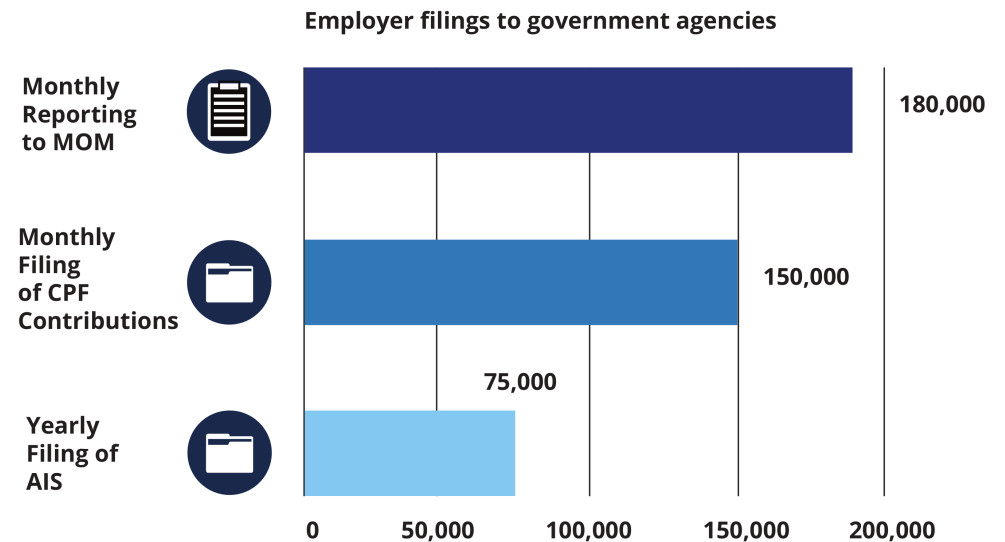


# Success story

Streamlining Payroll Submissions in Singapore: APEX's  
Solution for Efficient Government Reporting

# The Problem

- Singapore-registered companies must regularly submit employee salary, workforce, and CPF contribution data to various government agencies (IRAS, MOM, CPFB).
- All required data already exists within companies' payroll or HR systems.
- The challenge is to integrate government submissions with these systems to streamline processes, reduce manual work, and save employers time and resources.



# The solution

- APEX unifies APIs from multiple government agencies on one secure platform.
- It enables developers, including HR and payroll software providers, to integrate easily with these services.
- APEX removes the need for manual filing reducing administrative burden and supports better compliance with payroll-related regulations.

